The Gas Laws

Answer key with worked solutions

A. Understanding

1. Particles and pressure.

> Pressure comes from gas particles colliding with the container walls. In a smaller volume the same particles hit the walls **more often**, so the pressure is higher.

2. Why kelvin?

> Volume and pressure are proportional to the **absolute** temperature, which starts at absolute zero (0 K). The Celsius scale starts at an arbitrary point, so ratios of Celsius temperatures are meaningless (and 0 °C would mean dividing by zero).

B. Simple gas laws

3. Boyle.

> $$V_2 = \frac{P_1 V_1}{P_2} = \frac{(100\ \text{kPa})(5.0\ \text{L})}{250\ \text{kPa}} = \mathbf{2.0\ L}$$
>
> The units of kPa cancel, leaving L.

4. Charles.

> $T_1 = 20\ ^\circ\text{C} + 273.15 = 293.15\ \text{K}$; $T_2 = -30\ ^\circ\text{C} + 273.15 = 243.15\ \text{K}$
>
> $$V_2 = V_1 \times \frac{T_2}{T_1} = 250\ \text{mL} \times \frac{243.15\ \text{K}}{293.15\ \text{K}} = \mathbf{207\ mL}$$
>
> The units of K cancel, leaving mL.

5. Gay-Lussac.

> $T_1 = 15\ ^\circ\text{C} + 273.15 = 288.15\ \text{K}$; $T_2 = 45\ ^\circ\text{C} + 273.15 = 318.15\ \text{K}$
>
> $$P_2 = P_1 \times \frac{T_2}{T_1} = 220\ \text{kPa} \times \frac{318.15\ \text{K}}{288.15\ \text{K}} = \mathbf{243\ kPa}$$
>
> The units of K cancel, leaving kPa.

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C. Combined gas law

6. Combined.

$T_1 = 25\ ^\circ\text{C} + 273.15 = 298.15\ \text{K}$; $T_2 = 100\ ^\circ\text{C} + 273.15 = 373.15\ \text{K}$

$$\begin{aligned} V_2 &= \frac{P_1 V_1 T_2}{P_2 T_1} \\ &= \frac{(1.00\ \text{atm})(4.00\ \text{L})(373.15\ \text{K})}{(2.50\ \text{atm})(298.15\ \text{K})} \\ &= \mathbf{2.00\ L} \end{aligned}$$

The units of atm and K cancel, leaving L.

D. Ideal gas law

7. Pressure.

Volume is in L and the answer is wanted in atm, so use $R = 0.08206\ \text{L·atm/(mol·K)}$

$$\begin{aligned} P &= \frac{nRT}{V} \\ &= \frac{(0.250\ \text{mol})\left(0.08206\ \frac{\text{L·atm}}{\text{mol·K}}\right)(300\ \text{K})}{5.00\ \text{L}} \\ &= \mathbf{1.23\ atm} \end{aligned}$$

The units of mol, K and L cancel, leaving atm.

8. Carbon dioxide.

(a) $T = 27\ ^\circ\text{C} + 273.15 = 300.15\ \text{K}$

Pressure is in atm, so use $R = 0.08206\ \text{L·atm/(mol·K)}$

$$\begin{aligned} n &= \frac{PV}{RT} \\ &= \frac{(1.50\ \text{atm})(2.00\ \text{L})}{\left(0.08206\ \frac{\text{L·atm}}{\text{mol·K}}\right)(300.15\ \text{K})} \\ &= \mathbf{0.122\ mol} \end{aligned}$$

The units of atm, L and K cancel, leaving mol.

(b) $M(\text{CO}_2) = 12.01\ \text{g/mol} + 2(16.00\ \text{g/mol}) = 44.01\ \text{g/mol}$

$$m = n \times M = 0.12181\ \text{mol} \times 44.01\ \text{g/mol} = \mathbf{5.36\ g}$$

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E. Challenge

9. Molar mass.

$T = 0\ ^\circ\text{C} + 273.15 = 273.15\ \text{K}$

Pressure is in atm, so use $R = 0.08206\ \text{L·atm/(mol·K)}$

$$n = \frac{PV}{RT}$$

$$= \frac{(1.00\ \text{atm})(0.560\ \text{L})}{\left(0.08206\ \frac{\text{L·atm}}{\text{mol·K}}\right)(273.15\ \text{K})}$$

$$= 0.02498\ \text{mol}$$

$$M = \frac{m}{n} = \frac{1.00\ \text{g}}{0.02498\ \text{mol}} = \mathbf{40.0\ g/mol}$$

The gas is **argon** (39.95 g/mol).

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